

Euclid's Elements — Book II

Proposition 3

Formal Reconstruction — Technical Documentation

Jurek Róžański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJTEAMLAB

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.

1 Introduction

Proposition II.3 is the second immediate reuse of the rectangle calculus established in Proposition II.1. Its classical form is again a statement about a straight line cut at an arbitrary point, but now one of the two resulting segments plays a distinguished role. The rectangle contained by the whole line and that segment is decomposed into a rectangle contained by the two parts and the square on the distinguished segment.

The current Lean reconstruction makes this structure unusually transparent. Its only direct import is the II.1 module:

```
import CGJteamLab.Proposition2_1
```

The proof first recognizes the supplied square on MC as a rectangle contained by MC, MC , and then applies the binary form of II.1 with the fixed segment specialized to MC . No new dissection theorem, side-order swap, or proposition-local helper theorem is needed.

This is an important contrast with II.2. There the Euclidean order of the two containing segments did not match the order produced by the binary II.1 theorem, so cyclic rectangle rotation was required. In II.3 the order already matches:

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Rect}(MC, MC).$$

The square on MC is then simply a chosen representative of the final $\text{Rect}(MC, MC)$ term.

Thus II.3 is not formalized through multiplication of segment lengths or through a numerical area function. Its apparently algebraic shape is realized by an exact geometric cut together with representative independence in the rectangle API.

2 Euclid’s Statement and Classical Configuration

2.1 The statement

In Heath’s wording, Proposition II.3 states:

If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.

If AB is cut at C and the distinguished part is BC , the geometric identity is

$$\text{Rect}(AB, BC) = \text{Rect}(AC, CB) + \text{Sq}(BC).$$

As throughout the Book II documentation, Rect and Sq are expository shorthand for geometric figures and their content. They are not number-valued functions.

2.2 The classical diagram

Euclid constructs the square $CDEB$ on the distinguished segment CB by Proposition I.46. The side ED is produced, and through A a line is drawn parallel to the corresponding sides of the square by Proposition I.31 [?]. The resulting larger rectangle has adjacent side magnitudes AB and BC . The line through C separates it into a rectangle with side magnitudes AC, CB and the square on CB .

The classical state is shown first in its own lettering. The square $CDEB$ on CB is already one of the two final pieces, while the parallel through A extends it to the whole rectangle on AB and BC .

The current Lean lettering is related to the classical roles by

classical role	Lean role
AB	BC
C	M
AC	BM
CB	MC

so the formal cut relation is $B - M - C$ and the distinguished segment is MC . The configured outer rectangle is $BCED$, with $CE \cong MC$, while the cut through M meets the opposite side at L . That supplied Lean configuration is shown later, exactly where the II.1 binary split is invoked.

Diagrammatic audit. Two pictures are sufficient for II.3. Figure 1 records the source construction and its immediate rectangle-plus-square partition. The second picture is deferred to the formal proof step because it contains additional interface data—in particular the diagonal intersection witness X and a square representative that need not be the literal internal cut piece. Recognizing that supplied square as a rectangle contained by MC, MC does not change the geometry, so it does not merit a separate snapshot.

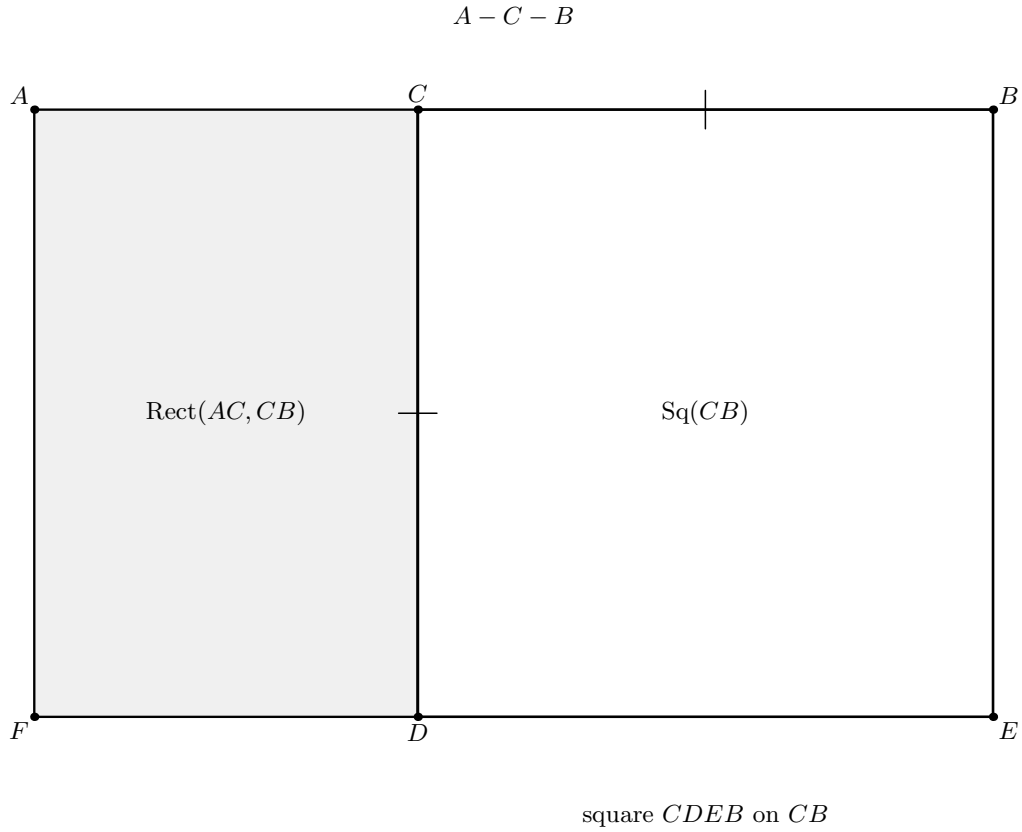


Figure 1: The classical configuration of Euclid II.3. The line is cut as $A - C - B$. The square $CDEB$ is erected on the distinguished part CB , and its lower side is produced to F while AF is drawn parallel to the square sides. The resulting whole rectangle $ABEF$ is visibly the union of the rectangle contained by AC, CB and the square on CB . No diagonal or scissors-refinement witness is shown, because those belong to the present formal interface rather than to Euclid's local construction.

3 Classical Proof

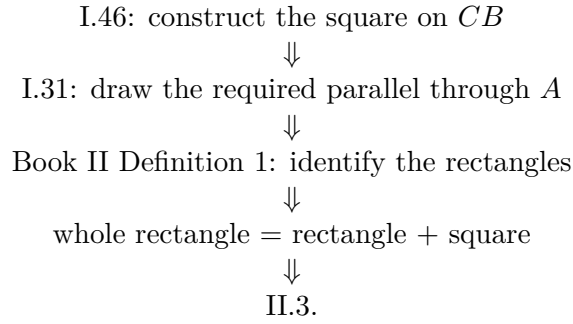
Let AB be cut at C , and choose BC as the segment named in the proposition. Construct the square $CDEB$ on CB by I.46. Produce one side of the square and through A draw a line parallel to the opposite sides by I.31. This forms a larger rectangle whose adjacent side magnitudes are AB and BC .

The line through C partitions that larger rectangle into two pieces. The first has adjacent side magnitudes AC and CB , and the second is precisely the square on CB . Hence

$$\text{Rect}(AB, BC) = \text{Rect}(AC, CB) + \text{Sq}(BC).$$

This is Euclid's entire content argument: the whole rectangle is constituted by the two displayed subfigures.

The local classical dependency graph is therefore



Euclid does not cite Proposition II.1 in this proof. The current Lean proof, by contrast, is explicitly organized as a corollary of the reusable II.1 binary theorem. The classical and formal dependency graphs therefore prove the same geometric identity through different library organizations.

4 The Geometric Identity

In the current Lean lettering, $B - M - C$. The theorem fixes the segment MC and applies it to the whole BC and the two parts BM, MC . Its exact geometric content is

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Sq}(MC).$$

A modern algebraic shadow is obtained by writing

$$x = y + z,$$

where x corresponds to BC , y to BM , and z to MC . Then the pattern is

$$xz = yz + z^2,$$

or equivalently

$$(y + z)z = yz + z^2.$$

This formula is useful only as a mnemonic. The Lean proof never forms the products xz or yz as segment-arithmetic objects. The actual objects are rectangle representatives and a square, related by exact scissors equality.

5 Reusable Book II Infrastructure

The current file begins with

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction.

5.1 The binary form of II.1

The reusable theorem is

```
euclid_proposition_2_1_two
```

with geometric content

$$\text{Rect}(BC, PQ) \simeq_{\text{sc}} \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

whenever $B - M - C$ and the required rectangular cut data are supplied.

For II.3 the fixed segment is specialized directly to

$$PQ = MC.$$

Therefore II.1 returns

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Rect}(MC, MC),$$

which already has exactly the containing-side order required by the public II.3 theorem.

5.2 A square as $\text{Rect}(MC, MC)$

The square hypothesis

```
hSquare : IsSquare Geo M C F G
```

is converted by

```
square_is_rectangle_contained_by_side
```

into

```
IsRectangleContainedBy Geo M C F G M C M C
```

so the supplied square is a concrete representative of the final $\text{Rect}(MC, MC)$ term.

5.3 What II.3 does not use

Unlike II.2, the current proof does not call

```
rectangle_contained_by_swap  
rectangle_contained_by_comm
```

The absence is mathematically informative. The binary II.1 theorem already places BM and MC first and the fixed segment MC second, exactly as II.3 needs. No geometric analogue of exchanging factors is required.

6 Proof Architecture of the Reconstruction

The public proof has only two mathematical stages.

6.1 Step 1: recognize the square

The first local fact is

```

have hSquareContained :
  IsRectangleContainedBy Geo
    M C F G M C M C :=
  square_is_rectangle_contained_by_side
    Geo M C F G hSquare

```

This moves from the square-specific geometric predicate to the general “rectangle contained by” language used by II.1.

Figure 2 records the configured cut that the next step will pass to the binary II.1 theorem.

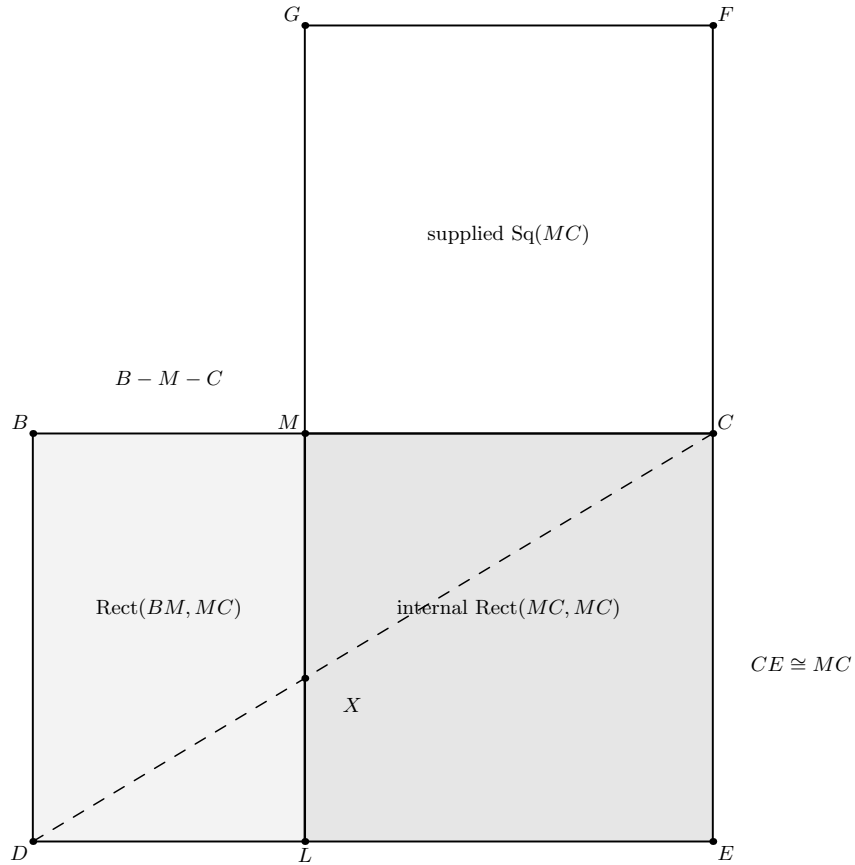


Figure 2: The configured geometric state consumed by the single call to `euclid_proposition_2_1_two`. The outer rectangle $BCED$ is cut at $B - M - C$ by ML , producing a left piece representing $\text{Rect}(BM, MC)$ and an internal right piece representing $\text{Rect}(MC, MC)$. The congruence $CE \cong MC$ provides the fixed second-side magnitude. The dashed diagonal DC meets the cut at X , the hidden exact-dissection witness inherited from II.1. The square $MCFG$ is deliberately drawn separately: Step 1 certifies it as another representative of $\text{Rect}(MC, MC)$, so the public conclusion need not reuse the literal internal piece.

6.2 Step 2: instantiate II.1 with the fixed segment MC

The remainder of the proof is one direct call:

```
exact
  euclid_proposition_2_1_two
    Geo
    B C D E L M X
    M C
    hRect
    hCE_MC
    hBMC
    hDLE
    hDXC
    hMXL
    hLeftPar
    hRightPar
    U0 U1 U2 U3
    V0 V1 V2 V3
    M C F G
    hWhole
    hProduct
    hSquareContained
```

Diagrammatic audit. Figure 2 is the only genuinely new formal state. Step 1 changes only the predicate under which the already supplied square is read; Step 2 is one binary II.1 dissection. After that call there is no local transitivity chain, summand rearrangement, side-order rotation, cancellation, or additional construction. A third figure would therefore depict only a change of representation or notation rather than new geometry.

No scissors transitivity, addition, symmetry, or representative rotation is performed locally in II.3. Those mechanisms have already been packaged inside the representative-independent binary theorem of II.1.

This makes the formal DAG exceptionally short:

$$\begin{array}{c}
 \text{square } MCFG \\
 \Downarrow \\
 \text{square_is_rectangle_contained_by_side} \\
 \Downarrow \\
 \text{Rect}(MC, MC) \text{ representative} \\
 \Downarrow \\
 \text{euclid_proposition_2_1_two with } PQ = MC \\
 \Downarrow \\
 \text{euclid_proposition_2_3.}
 \end{array}$$

7 Lean Formalization

The current public theorem is:

```
theorem euclid_proposition_2_3
  [HilbertIncidence Geo]
```

```

[HilbertEuclideanPlane Geo]

(B C D E L M X : Geo.Point)
(hRect : IsRectangle Geo B C E D)
(hCE_MC : Geo.Congruent C E M C)

(hBMC : Geo.Between B M C)
(hDLE : Geo.Between D L E)
(hDXC : Geo.Between D X C)
(hMXL : Geo.Between M X L)
(hLeftPar :
  IsParallelogram Geo L D B M)
(hRightPar :
  IsParallelogram Geo C E L M)

(F G : Geo.Point)
(hSquare : IsSquare Geo M C F G)

(U0 U1 U2 U3 : Geo.Point)
(V0 V1 V2 V3 : Geo.Point)

(hWhole :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 B C M C)

(hProduct :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B M M C) :
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo U0 U1 U2 U3)
  (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
    hilbertParallelogramTerm Geo M C F G) := by

```

Several features of this signature are mathematically significant.

First, the theorem assumes a configured outer rectangle and its cut. It does not construct that diagram from the bare data $B - M - C$. Euclid performs the square and parallel constructions in his proof; the formal theorem receives the corresponding geometry as explicit hypotheses.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of

$$\text{Rect}(BC, MC) \quad \text{and} \quad \text{Rect}(BM, MC).$$

The conclusion is therefore not restricted to the literal pieces drawn in the configured cut.

Third, the square $MCFG$ is concrete, but the II.1 representative-independence machinery permits it to stand for the $\text{Rect}(MC, MC)$ piece even when it is not the internal right-hand rectangle of the supplied cut diagram.

Fourth, the conclusion is exactly

```
HilbertScissorsEq
```

rather than the more general relation

```
HilbertScissorsEquicomplementable
```

and it is not equality of real numbers.

8 Relationship with Earlier Book II Propositions

8.1 Proposition II.1

II.3 is a direct formal corollary of II.1. The import is

```
import CGJteamLab.Proposition2_1
```

and the only substantial theorem called in the final proof is

```
euclid_proposition_2_1_two
```

The specialization $PQ = MC$ converts the general binary distributive pattern into II.3 immediately.

The relationship can be written schematically as

$$\text{Rect}(BC, PQ) = \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

followed by

$$PQ := MC.$$

The final term then becomes $\text{Rect}(MC, MC)$, represented by the square on MC .

8.2 Proposition II.2

The current Lean file does *not* import Proposition II.2, and its public theorem is not used in the proof. II.2 and II.3 are therefore sibling corollaries of the II.1 binary API rather than a formal chain $\text{II.1} \rightarrow \text{II.2} \rightarrow \text{II.3}$.

The contrast is nevertheless useful. II.2 required cyclic representative rotation because its chosen side order was opposite to the order returned by II.1. II.3 chooses the fixed segment MC in precisely the position expected by II.1, so even that transport disappears. This is evidence that the II.1 API is not merely sufficient but well aligned with the geometry of the next Book II identities.

9 Relationship with Book I Infrastructure

Euclid's local proof explicitly calls I.46 to construct the square and I.31 to draw the parallel. The current Lean theorem does not replay those constructions. Instead it assumes the resulting rectangle, cut, parallelograms, side congruence, and square as data.

Below II.1, however, the formal dependency chain still rests on the Book I infrastructure for rectangles, right angles, squares, parallelograms, and exact scissors decomposition. The binary split ultimately uses the exact rectangle cut machinery developed over the Book I scissors layer. Thus the difference is one of modular organization, not a change from synthetic to numerical geometry.

10 Formalization Notes and Hidden Geometric Data

The elementary picture hides several obligations that appear explicitly in the current theorem signature:

- $B - M - C$, fixing the cut point and its orientation on the whole side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the diagonal and the cut line used by the inherited exact scissors theorem;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- $CE \cong MC$, making the outer rectangle have the required fixed second side magnitude;
- a supplied square $MCFG$;
- arbitrary representatives of the whole rectangle and the BM -by- MC product rectangle.

The point X is invisible in Euclid’s verbal statement. It belongs to the triangulated exact-dissection interface inherited from II.1. It should therefore be read as hidden formal geometry rather than as an additional step in Euclid’s classical construction.

There are also useful negative observations. No `SameRay`, `SameSide`, or `OppositeSide` hypothesis occurs in `Proposition2_3.lean` itself, and no side-order swap is performed. Those lower-level obligations have already been absorbed by the reusable rectangle and II.1 layers.

11 Geometry, Content, Representation, and Later Arithmetic

11.1 Geometry

The geometric layer consists of the rectangle $BCED$, the cut point M , the opposite cut point L , the diagonal witness X , the two parallelogram pieces, the congruence $CE \cong MC$, and the square $MCFG$.

11.2 Content / scissors

The conclusion is the exact relation `HilbertScissorsEq`. The public proof does not weaken the result to general equicomplementability. Exactness is inherited from the binary rectangle split and representative transports already proved in II.1.

11.3 Representation

Representation is present but simpler than in II.2. The theorem accepts an arbitrary rectangle representative for $\text{Rect}(BC, MC)$ and an arbitrary representative for $\text{Rect}(BM, MC)$, together with a concrete square on MC . The binary II.1 theorem is already independent of the particular concrete rectangles appearing in the cut, so no additional representation transport is required locally.

11.4 Later arithmetic / numerical area

Neither layer is used. No segment product is defined, no fixed unit segment is introduced, and no real-valued area function appears. The expression $xz = yz + z^2$ is only the modern algebraic shadow of a geometric scissors identity.

12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore lives in the same Euclidean project layer as II.1 and II.2. Inspection of the current 96-line source shows one public theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as “axiom-free.” The result inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output for II.3 was supplied with the documentation sources, so a transitive axiom list is not asserted here.

The implementation checkpoint records that the II.3 Lean theorem had already been build-checked during formalization. The present documentation pass uses the current source as authoritative but does not contain the full Lean project needed to repeat that build locally.

13 Hartshorne’s Semantic Control

Hartshorne’s general analysis of Euclid’s area theory states that the results of Book II are valid when Euclid’s equality of figures is read as equality of content [?]. His condensed list of selected Book II propositions does not separately single out II.3, so the source should not be cited as providing a special proof of this proposition.

The general semantic point is sufficient. II.3 compares a rectangle with the union of another rectangle and a square. In the present reconstruction this comparison is sharpened, for this configured case, to exact scissors equality. The modern identity

$$xz = yz + z^2$$

therefore belongs to Hartshorne’s “algebra of areas” only as an interpretive shadow of operations on geometric figures.

II.3 also clarifies the representative issue. The term called “the square on MC ” need not be the literal internal square-shaped piece of the cut; the formal theorem accepts the separately supplied square $MCFG$ because the rectangle calculus already proves independence of representatives at the scissors level.

14 Hilbert's Layering

Hilbert provides the complementary warning against reading later segment arithmetic into the proof. His synthetic theory of plane content distinguishes finite equidecomposition from the more general content relations, while his arithmetic of segments is introduced later and requires a fixed unit segment and additional parallel constructions [?].

II.3 remains entirely below that arithmetic boundary. Its formal chain is

$$\begin{array}{c} \text{rectangle and square geometry} \\ \Downarrow \\ \text{exact binary dissection from II.1} \\ \Downarrow \\ \text{representative-independent exact scissors equality.} \end{array}$$

In fact II.3 is even more direct than II.2: no geometric commutativity theorem is needed because the ordered containing segments already agree with the II.1 binary interface.

15 Dependency Summary

The actual Lean dependency chain is

$$\begin{array}{c} \text{Hilbert incidence + Euclidean plane layer} \\ \Downarrow \\ \text{Book I right-angle, square, parallelogram, and scissors infrastructure} \\ \Downarrow \\ \text{HilbertRectangle} \\ \text{IsRectangleContainedBy} \\ \text{square_is_rectangle_contained_by_side} \\ \Downarrow \\ \text{Proposition II.1 binary API} \\ \text{euclid_proposition_2_1_two} \\ \text{specialized by } PQ = MC \\ \Downarrow \\ \text{euclid_proposition_2_3.} \end{array}$$

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{rectangle cut into rectangle + square} \longrightarrow \text{II.3.}$$

New geometric axiom in II.3:	no
New content axiom in II.3:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New <code>HilbertRectangle</code> theorem introduced by II.3:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Imports II.2:	no
Direct side-order transport used:	no
Exact <code>HilbertScissorsEq</code> :	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local <code>sorry/admit</code> :	no

16 Conclusion

Proposition II.3 is one of the cleanest tests of the Book II rectangle API. Classically, Euclid constructs a square on one segment and enlarges it by a parallel construction until the whole rectangle contained by the original line and that segment is obtained. Formally, the same identity is already contained in the binary theorem of II.1.

The decisive specialization is simply

$$PQ = MC.$$

It turns the second part rectangle of II.1 into $\text{Rect}(MC, MC)$, and the supplied square on MC provides exactly that representative. Nothing else is needed.

This also explains the precise relation between II.2 and II.3. Both are formal corollaries of II.1, but II.2 needs geometric side-order transport while II.3 does not. The result is therefore a very short Lean proof without being an algebraic proof in the modern numerical sense: the content remains rectangles, squares, congruent sides, and exact scissors equality throughout.